

- 5 (a) A student decides to build a temperature probe and sets up the circuit using a thermistor and a $5.00\text{ k}\Omega$ resistor as shown in Fig. 5.1. The battery has e.m.f. 9.00 V and negligible resistance.

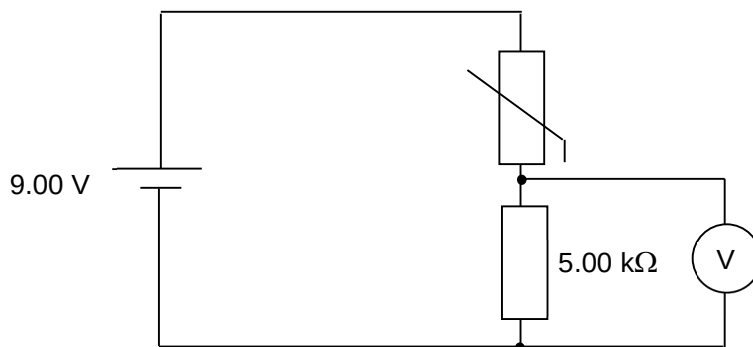


Fig. 5.1

- (i) When the thermistor is at 30.0°C , the thermistor's resistance is $1.25\text{ k}\Omega$.

Calculate the reading on the voltmeter at this temperature.

voltmeter reading = V [1]

- (ii) The student removed the thermistor and voltmeter from the circuit. He added in a light dependent resistor (LDR) and a lamp to build a lighting system which switches on automatically when the surrounding gets dark.

In the space provided below, draw a circuit diagram to show how the electrical components can be connected.

[2]

- (b) Fig. 5.2 shows a circuit which may be used to compare the resistance R of an unknown resistor with a $100\ \Omega$ resistor. The distances l from one end of the potentiometer slide-wire to the balance point are 400 mm and 588 mm when X is connected to Y and to Z respectively. The length of the slide-wire is 1.000 m.

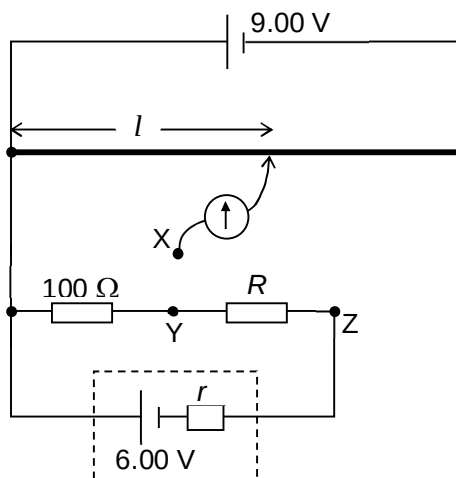


Fig. 5.2

Determine

- (i) the value of resistance R ,

$$R = \dots\dots\dots \Omega \quad [2]$$

- (ii) the value of internal resistance r .

$$r = \dots\dots\dots \Omega \quad [2]$$

6 (a) By reference to the photoelectric effect state what is meant by the work function of a